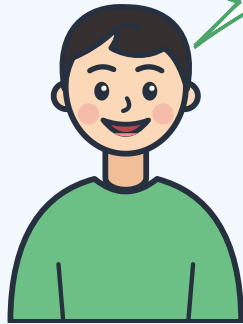


10. PROTOCOLS: AGREED RULES

A CONVERSATION ANALOGY

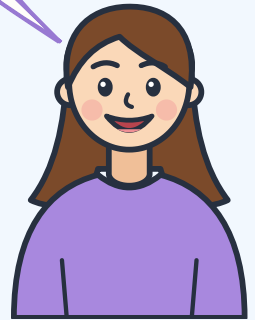


Person A

Shall we meet at 3 pm?

We both understand because we agree on rules:

- use the same language (syntax)
- know what the words mean (meaning)
- take turns in the right order (order)
- ask for a repeat if not understood (error handling)



Person B

Yes, at 3 pm!

Protocols let independent devices interpret the same exchange.

A LAYERED PROTOCOL STACK

Higher layers
(closer to users)

Application

— network applications
(e.g. HTTP, DNS)

Transport

— end-to-end
communication (e.g.
TCP, UDP)

Internet

— addressing and routing
(e.g. IP)

Link

— local delivery on a
network (e.g. Ethernet)

Physical

— bits over a medium
(wires, radio, fibre)

Lower layers
(closer
to hardware)

KEY IDEAS



SYNTAX

Format of the data. E.g. message fields, headers, character encoding.



MEANING

What the data represents. E.g. `GET` means "request a web page".

1 2 3

ORDER

Steps must happen in a specific sequence. E.g. connect before sending data.



ERROR HANDLING

Detect problems and recover. E.g. checksums, timeouts, retransmission, error messages.

EXAMPLES OF COMMON PROTOCOLS

Ethernet

(link layer)



Moves frames on
a local network.

IP

(internet layer)



Addresses
packets and
helps route them
between networks.

TCP

(transport layer)



Reliable, ordered
delivery with
error handling
(e.g. retransmits
lost data).

UDP

(transport layer)



Simple, best-effort
delivery. No
guarantee of
delivery, ordering
or recovery.

DNS

(application layer)



Translates names
(e.g. example.com)
to IP addresses.

HTTP

(application layer)



Transfers web
content between
browsers and
web servers.



REMEMBER: Protocols are agreements. Without them, devices could not understand each other.